Richard Coley
Unix / Open Systems Community Contributions

Documented chronology, 1992-2000 (with unresolved archival leads)

Scope. This chronology consolidates the evidence recovered so far from Eric S. Raymond's "Known Bugs in the USL UNIX Distribution", XFree86 changelogs and community material, lsof release history, and the ongoing search for GCC and Samba contributions. Items are labelled by evidence strength so that recollection is kept separate from documentary confirmation.

Evidence labels: CONFIRMED = directly supported by surviving project/archive material; STRONGLY IDENTIFIED = matching project/date/technical context but incomplete attribution; RECOLLECTED / UNRESOLVED = remembered contribution not yet tied to the original archival record.

Chronological record

3 November 1992 - USL / SVR4 - BSD compatibility signal bug (Eric S. Raymond's usl-bugs v7.0)

CONFIRMED

- Eric S. Raymond's Version 7.0 of "Known Bugs in the USL UNIX Distribution" contains a development-tools section on broken BSD/UCB compatibility libraries in USL SVR4.
- The document introduces a contribution with the line: "Here's more on the signals problem, thanks to Richard".
- The report demonstrates that a program calling signal(SIGPIPE, SIG_IGN) and linking with the BSD compatibility library can still be killed by SIGPIPE.
- The diagnosis identifies a missing 'else' clause in the compatibility-library signal() implementation.
- The practical result is that a signal handler can be changed repeatedly to another function handler, but attempts to set the handler to SIG_DFL or SIG_IGN fail.
- A practical workaround is supplied: install a no-op function handler rather than SIG_IGN.
- The document explicitly notes that the bug caused X11R5 problems, including the X server dying when a client closed. Christoph Badura later confirmed the defect and sent USL a source fix.
- This establishes an early pattern that recurs later: SVR4 portability, BSD/UCB compatibility, low-level API diagnosis, and feeding reproducible fixes/workarounds back to the Unix/X11 community.

Evidence: Uploaded usl-bugs Version 7.0; signal section in Development Tools Problems.

1992-1994 (exact dates still to recover) - GCC / GNU compiler portability and Unix porting community

RECOLLECTED / UNRESOLVED

- Richard recalls posting about GCC porting and portability issues while working in Pyramid/Nixdorf/Siemens Unix environments.
- The technical setting is consistent with the 1992 usl-bugs document, which discusses failures caused by mixing System V headers with BSD/UCB compatibility libraries and specifically mentions GNU C compiler builds.
- Known historical identity variants now worth searching include rc, rcoley, r.coley, rich and richard, together with pyra.co.uk, pyramid.com, sni.co.uk, siemens-nixdorf.co.uk and local-host style identities such as richard@pyrcent.

- No individual GCC post has yet been recovered strongly enough to quote or date, so this remains deliberately separate from the confirmed XFree86 GCC/Solaris fix in 2000.

Evidence: User recollection plus period technical context; original GNU/GCC messages still to recover.

15 September 1994 - XFree86 / comp.os.linux.x - IDEK/Iiyama Vision Master 17 monitor configuration

CONFIRMED

- A surviving XFree86-era community record attributes monitor timing/configuration information to Richard Coley at a Pyramid-era address.
- The contribution covered the IDEK/Iiyama Vision Master 17 (MF8617) monitor.
- This is a community-support contribution rather than a source-code patch: working monitor timings/modelines were valuable in the XFree86 3.x era because incorrect timings could produce an unusable display and users frequently relied on shared known-good configurations.
- It shows active participation in the XFree86 community by 1994, preceding the later code-level SVR4 fixes.

Evidence: Surviving Usenet/XFree86 monitor configuration material.

20 November 1996 - XFree86 3.2b - DGA framebuffer mapping for SVR4

CONFIRMED

- XFree86 CHANGELOG entry: "Fix DGA fb mapping for SVR4 (Richard Coley)."
- This is almost certainly the contribution Richard remembered as the XFree86 3.x SVR4 mmap/framebuffer-mapping fix.
- DGA (Direct Graphics Access) required mapping framebuffer memory into a process/client address space. On SVR4, access to physical/framebuffer memory depended on OS-specific mapping semantics and device interfaces.
- The change therefore sits at a low-level OS/hardware boundary: XFree86's direct framebuffer access had to be adapted to SVR4's memory-mapping behaviour.
- Earlier investigation of Matrox MGA memory-probing issues was relevant background but was not the best description of Richard's own contribution; the changelog identifies his fix specifically as DGA framebuffer mapping on SVR4.

Evidence: XFree86 CHANGELOG supplied in this conversation.

14 April 1997 - XFree86 3.2At - loader fixes on SVR4.0.4 (#277)

CONFIRMED

- XFree86 CHANGELOG entry: "Fix loader under SVR4.0.4 (lstat,fstat,mknod) (#277, Richard Coley)."
- This was a portability fix in the XFree86 runtime/module loader rather than a graphics-driver change.
- The APIs named in the entry - lstat(), fstat() and mknod() - point to differences in filesystem/device handling encountered on SVR4.0.4.
- The issue number (#277) is a useful archival search key for any surviving bug database, developer mail or patch correspondence.
- Together with the 1996 framebuffer-mapping fix, it demonstrates repeated work making XFree86 function correctly on System V Release 4 rather than only on the more common BSD/Linux platforms.

Evidence: XFree86 CHANGELOG supplied in this conversation.

24 June 1997 - lsof 4.12 - Pyramid Reliant UNIX, OBJFS and Nile/Jolt support

CONFIRMED

- The lsof project history explicitly credits Richard Coley <rcoley@pyra.co.uk>.
- Richard helped make Pyramid OBJFS handling conditional on the appropriate platform/header support.
- He also helped correct MkKernOpts so that the proper compile-time definitions were generated for the Pyramid Nile/Jolt architecture.
- This is deep portability work: lsof depends on kernel structures, vnode/filesystem internals, device information and platform-specific kernel interfaces.
- The explicit pyra.co.uk address provides one of the strongest identity anchors for searches of contemporary archives.

Evidence: lsof 4.12 project/release history.

6 July 1997 - XFree86 3.9i - mouse/valuator initialisation across OS-specific MouseProc paths

CONFIRMED

- The XFree86 CHANGELOG credits Holger Veit and Richard Coley with a fix for missing valuator initialisation in OS-specific MouseProc() implementations.
- The affected paths included SCO, OS/2 and Xqueue.
- In the X input model, valuators represent axis data; missing or incorrect initialisation could produce broken or unpredictable pointer behaviour.
- This is significant because it is not narrowly Pyramid-specific. It shows participation in a general multi-platform XFree86 input portability problem spanning several operating-system backends.

Evidence: XFree86 CHANGELOG supplied in this conversation.

9 July 1997 - lsof 4.13 - Pyramid ttyfs, FIFO handling and UCB compatibility cleanup

CONFIRMED

- Richard helped lsof recognise and handle Pyramid ttyfs vnodes.
- The same release history records Pyramid FIFO-related corrections.
- Use of the Pyramid UCB compatibility library was eliminated at Richard's suggestion.
- That last point is particularly consistent with the 1992 USL signal work: both involve the practical hazards of BSD/UCB compatibility layers on System V-derived Unix systems.
- The 1992 and 1997 records therefore form a coherent technical thread rather than unrelated contributions.

Evidence: lsof 4.13 project/release history.

1997-1999 (exact dates still to recover) - Further GCC / Unix portability activity

RECOLLECTED / UNRESOLVED

- Richard recalls continuing to contribute to GCC/portability discussions.
- The confirmed work around this period - XFree86 SVR4 loader fixes, lsof kernel-portability work and UCB compatibility cleanup - matches the kinds of issues exposed by compiler/toolchain ports on proprietary Unix systems.
- The archival search needs to target old GNU mailing-list mirrors and Usenet using all known identity variants rather than only rcoley@pyra.co.uk.

Evidence: User recollection; individual GCC messages not yet recovered.

c. 1998-1999 - Samba 2.x - maximum number of open files / file-descriptor limits (CSFB)

STRONGLY IDENTIFIED, ATTRIBUTION STILL TO RECOVER

- Richard recalls contributing a Samba 2.x fix while working at csfb.com, concerning the maximum number of open files available to Samba/smbd.
- Samba 2.0 introduced a "max open files" configuration parameter and code around Unix process file-descriptor limits.
- The relevant implementation area involves set_maxfiles(), RLIMIT_NOFILE/getrlimit()/setrlimit(), OPEN_MAX/ulimit-style limits and the interaction with Samba's desired descriptor count.
- A later Samba 2.0.7-era Solaris discussion confirms that this was a real and important design area: smbd could be constrained by the operating system's descriptor ceiling and required explicit resource-limit handling.
- That later 2001 Solaris patch was by another contributor, so it should not be attributed to Richard. The current target is the earlier Samba 2.0 introduction or correction that Richard remembers contributing from CSFB.
- Search keys for archival recovery: Richard Coley, rcoley, rc, r.coley, rich, csfb.com, max open files, set_maxfiles, RLIMIT_NOFILE, getrlimit, setrlimit, OPEN_MAX, ulimit and too many open files.

Evidence: User recollection plus Samba 2.0 documentation confirming the feature area; exact original patch/message still to recover.

9 February 2000 - XFree86 3.9.17d - Solaris 8 build fixes with GCC (#3615)

CONFIRMED

- XFree86 CHANGELOG entry: "Fix some build problems on Solaris 8 with gcc (#3615, Richard Coley)."
- This is the strongest recovered direct link between Richard's XFree86 work and GCC.
- The issue concerns toolchain/platform portability: building XFree86 on Solaris 8 with GCC rather than relying solely on Sun's native compiler environment.
- Issue #3615 provides another precise search key for surviving bug/patch archives.
- The entry continues the same portability theme seen in the earlier SVR4 work, but now on Solaris 8.

Evidence: XFree86 CHANGELOG supplied in this conversation.

2 August 2000 - XFree86 4.0.1a - Solaris 8 text-mode colourmap save/restore (#A.104)

CONFIRMED

- XFree86 historical changelog credits Richard Coley with saving/restoring the text-mode colourmap on Solaris.
- The purpose was to fix a blank-screen problem after exiting the X server on Solaris 8.
- This is a display-state restoration issue at the boundary between the X server, VGA hardware state and Solaris console/virtual-terminal behaviour.
- When X changes graphics state and colour lookup tables, those values must be restored correctly when returning to text mode. Failure to restore the palette can leave the console apparently black even though the machine is still running.
- This represents the latest confirmed item in the chronology recovered so far.

Evidence: Surviving XFree86/X.Org historical changelog; change identifier #A.104.

Cross-cutting technical themes

System V / SVR4 portability: Repeated fixes around SVR4 APIs, memory mapping, loaders, signals and compatibility-layer behaviour.

BSD/UCB compatibility: The 1992 signal() defect and the 1997 lsof UCB-library removal show a recurring concern with subtle incompatibilities introduced by compatibility libraries.

X11 / XFree86 integration: Contributions span monitor configuration, direct framebuffer mapping, runtime loading, mouse/input initialisation, Solaris build portability and console colourmap restoration.

Low-level operating-system interfaces: signal(), framebuffer mapping, lstat(), fstat(), mknod(), vnode handling, FIFO handling, ttyfs, resource limits and display-state restoration all sit close to the kernel/OS boundary.

Compiler and toolchain portability: GCC is part of the remembered early porting work and appears directly in the confirmed Solaris 8 XFree86 build fix.

Upstreaming knowledge from proprietary Unix: A recurring role was helping upstream open-source projects understand and support commercial Unix variants that their core maintainers often could not test directly.

Known identity/address search keys

Likely local-parts: rc, rcoley, r.coley, rich, richard

Known / relevant domains and host-style identities: pyra.co.uk, pyramid.com, sni.co.uk, siemens-nixdorf.co.uk, csfb.com, richard@pyrcent

These variants matter because 1990s Usenet and mailing-list gateways often preserved local host-style addresses or changed address forms over time.

Outstanding archival targets

- Recover the original GCC/GNU compiler-porting posts and date them precisely.
- Locate the original XFree86 bug/patch records for #277 and #3615 and, if possible, the DGA SVR4 framebuffer patch discussion.
- Recover the original Samba 2.x CSFB-era max-open-files contribution and tie it to a csfb.com address.
- Search surviving XFree86 developer/patch mailing-list archives for all address variants.
- Search older Samba developer/CVS archives for max open files, set_maxfiles and RLIMIT_NOFILE.

Summary

The documentary record recovered so far spans at least November 1992 to August 2000. The confirmed material is not a collection of unrelated bug reports: it shows a consistent pattern of Unix portability and integration work across USL/SVR4, X11/XFree86, lsof and Solaris, with likely additional GCC and Samba contributions still to be fully recovered from historical archives. A recurring role was diagnosing platform-specific behaviour on commercial Unix systems, constructing reproducible fixes or workarounds, and feeding that knowledge back into open-source communities.